

A Stochastic Modeling Approach for Phase Transition-Induced Errors in GBAS under Ionospheric Scintillation

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BIOGRAPHYIES

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Dr. Sam Pullen is a senior researcher within the GNSS Laboratory at Stanford University, where he received his Ph.D. in Aeronautics and Astronautics in 1996. He has supported the FAA and other service providers in developing system concepts, technical requirements, integrity algorithms, and performance models for GBAS, SBAS, and other GNSS services and applications. He has also performed GNSS system design, application development, risk assessment, and legal support through his consultancy, Sam Pullen Consulting. He was awarded the ION Early Achievement Award in 1999 and became an ION Fellow in 2017.

Dr. Maria Caamano received a Master's degree in Telecommunications engineering from the University of Oviedo, Spain, in March 2015, and a PhD in Aerospace Science and Technology from the Polytechnic University of Catalonia (UPC), Spain, in July 2022. Since May 2015, she has been working as a research associate at the Institute of Communications and Navigation at the German Aerospace Center (DLR). Her work focuses on the development of dual-frequency multi-constellation Ground Based Augmentation Systems (GBAS).

ABSTRACT

Ionospheric scintillation can introduce diffractive phase transitions that corrupt carrier-smoothed pseudoranges in Ground-Based Augmentation Systems (GBAS), introducing unmodeled errors that are not accounted for in existing GBAS standards. Current GBAS error models assume negligible carrier phase noise contributions. However, phase transitions during scintillation propagate through carrier smoothing filters as step-like biases, potentially degrading system integrity. This paper employs physics-based scintillation simulation to generate controlled error samples and develops a novel stochastic modeling method to derive sigma overbounds for phase transition-induced errors under scintillation conditions. We apply a two-component power-law phase screen model to generate extensive scintillation data under controlled conditions across comprehensive parameter spaces. To address the discrete characteristics of phase transition-induced errors, we propose a stochastic modeling approach that represents phase transitions as Skellam-distributed point processes. This enables analytical derivation of sigma overbounds that statistically extrapolate to rare-event probability levels required for aviation integrity. The proposed approach overcomes the finite sample constraints of empirical CDF overbounding by providing more conservative bounds through statistical extrapolation. Results reveal that ionosphere-free (IFree) and divergence-free (DFree) smoothing modes employed in dual-frequency GBAS architectures amplify phase transition-induced errors. IFree errors approximately 3.5 times larger than L1 single-frequency mode, while the DFree smoothing exhibits 7 times amplification. Phase transition-induced errors increase with

longer smoothing time constants, with 600-second smoothing resulting in approximately 2.4 times larger errors than 100-second smoothing. This work proposes a methodology to derive sigma overbounds for phase transition-induced errors that may need to be incorporated into GBAS protection level calculations. The derived parameters enable more comprehensive integrity assessment for aviation operations under scintillation. Although partial mitigation may be achievable through the exclusion processes of existing GBAS integrity monitors or scintillation-dedicated monitors, a thorough investigation into their effectiveness against phase transition-induced errors is required.

I. INTRODUCTION

The Ground-Based Augmentation System (GBAS) provides local-area differential GNSS corrections for precision approach and landing operations. GBAS ground facilities process GNSS measurements and broadcast differential corrections along with integrity parameters that enable aircraft to compute protection levels to ensure navigation safety with extremely low probabilities (10^{-7} to 10^{-9}) (RTCA, 2019). The integrity framework relies on statistical error models that bound residual errors in carrier-smoothed pseudoranges, with standard deviations (e.g., σ_{pr_gnd} and σ_{pr_air}) defined by the Ground Accuracy Designator (GAD) and Airborne Accuracy Designator (AAD) models (McGraw et al., 2000). Current GBAS error models assume that carrier phase noise contributions to smoothed pseudoranges remain negligible. These models were developed and validated using extensive field testing in mid-latitude regions under nominal conditions. However, this fundamental assumption breaks down during ionospheric scintillation, which occurs with some frequency in low-latitude regions. This creates an unmodeled potential integrity risk for GBAS operations in these areas.

Ionospheric scintillation occurs when GNSS signals propagate through irregular electron density structures in the ionosphere, particularly the equatorial plasma bubbles (EPBs) common in equatorial regions. During moderate to strong scintillation events, rapid fluctuations in signal amplitude and phase can cause gradual full-cycle changes in the carrier phase measurements during continuous tracking without loss of lock. These events, known as phase transitions (or diffractive cycle slips), introduce sudden step-like errors into carrier phase measurements (Breitsch et al., 2020; Carrano et al., 2013; Gherm et al., 2011; Humphreys et al., 2010). Unlike conventional cycle slips caused by signal loss of lock, phase transitions occur due to the diffractive nature of scintillation and cannot be reliably detected using standard cycle slip detection algorithms (Breitsch et al., 2022). This becomes even more difficult at the 2 Hz GBAS sampling rate. When phase transitions occur, they introduce full-cycle phase errors (approximately 19 cm for L1 and 24 cm for L5) that enter the smoothing filter as step inputs. These step inputs disturb the carrier smoothing filter, creating transient errors that persist in the smoothed pseudoranges until the filter fully reconverges. The critical issue for GBAS is that phase transitions create systematic errors in smoothed pseudoranges that exceed current error model bounds. When multiple phase transitions occur before the smoothing filter converges, these errors accumulate, resulting in residual errors that can reach several meters during strong scintillation events (Sun et al., 2025b). Current GBAS error models do not account for these phase transition effects, creating a significant gap in the integrity framework for equatorial operations.

Next-generation GBAS architectures employ dual-frequency multi-constellation (DFMC) capabilities to enhance performance and support Category II/III operations (Murphy et al., 2021). These systems utilize advanced dual-frequency smoothing techniques including ionosphere-free (IFree) and divergence-free (DFree) combinations designed to cancel or mitigate ionospheric delay variations (Hwang et al., 1999). However, these dual-frequency combinations exhibit unexpected vulnerability to phase transition effects. The combination coefficients used in IFree and DFree smoothing amplify phase transition errors in L1 and L5 signals rather than mitigating them. During scintillation, phase transitions occur on both L1 and L5 but are temporally uncorrelated over a few seconds (Jiao et al., 2016), causing dual-frequency combinations to amplify rather than cancel these errors. This results in larger smoothed pseudorange errors compared to single-frequency operations, contradicting the expected performance improvements of DFMC architectures.

Our recent work (Sun et al., 2025b) confirmed through empirical analysis that phase transitions during scintillation introduce smoothed pseudorange residuals exceeding current GBAS error models, demonstrating the need for a new scintillation error term. However, developing scintillation-specific error models faces fundamental limitations. Phase transitions produce heavy-tailed error distributions where extreme tail behavior critically determines the sigma overbounds required for GBAS integrity at rare-event probability levels (10^{-7} to 10^{-9}). Characterization of these rare events requires extensive statistical sampling that real-world data cannot adequately provide. Scintillation in GNSS signals exhibits high temporal and spatial variability, providing only sparse sampling under specific conditions. Conventional empirical cumulative distribution function (CDF) overbounding (DeCleene, 2000; Blanch et al., 2018) cannot reliably characterize extreme tails from such limited datasets. Even extensive data collection campaigns cannot generate sufficient samples to reliably estimate error bounds at aviation integrity levels, as each specific scintillation condition provides insufficient occurrences to achieve the sample sizes necessary for statistical confidence at these extremely low probability levels.

To overcome the limitations of real-world data, this study employs physics-based scintillation simulation using a two-component power-law phase screen model (Jiao et al., 2018; Rino et al., 2018). This approach generates controlled scintillation samples across comprehensive parameter spaces, enabling systematic analysis of phase transition effects under statistically consistent conditions impossible to achieve with real-world data. The simulation provides two key advantages: it generates scintillation effects on L1/L5 frequencies by assuming the same ionospheric irregularity structure, ensuring physical consistency; and it enables controlled parameter studies across the complete scintillation parameter space required for comprehensive error model development. This systematic approach overcomes the sampling limitations inherent in empirical analyses.

While physics-based simulation generates controlled scintillation data, characterizing error distributions at GBAS integrity probability levels still requires sample sizes that exceed practical computational limits. To address this fundamental limitation, this study develops a novel stochastic modeling methodology that represents phase transitions as stochastic point processes (Breitsch et al., 2020). Building on this representation, the approach enables analytical derivation of probability distributions and statistical extrapolation to the rare-event probability levels required for GBAS integrity without requiring extensive simulation sample sizes. The resulting model-based approach provides conservative error bounds (σ_{scint}) that reliably characterize extreme tail behavior far beyond what sample-based methods can achieve.

This study develops the comprehensive error models accounting for phase transitions under ionospheric scintillation in GBAS operations. The specific objectives are: (1) implement controlled scintillation simulations to overcome real-world data limitations; (2) develop stochastic models of phase transitions to enable rare-event probability characterization, and (3) derive model-based error bounds (σ_{scint}) across GBAS smoothing modes and time constants. By combining physics-based simulation with analytical stochastic modeling, this work enables statistical characterization of phase transition-induced errors at aviation integrity probability levels, achieving what empirical approaches cannot accomplish due to finite sample limitations. Section 2 presents the scintillation simulation framework based on the two-component power-law phase screen model. Section 3 develops the model-based overbounding methodology through stochastic modeling of phase transitions. Section 4 analyzes the resulting error bounds across scintillation parameters and different smoothing configurations. Section 5 concludes with implications for GBAS integrity assurance under scintillation conditions.

II. SCINTILLATION SIMULATION FRAMEWORK

2.1 Two-Component Power-Law Phase Screen Model

The physics-based simulation framework employed in this study relies on the phase screen approximation, which replaces the complex interaction between radio waves and irregularity medium with an equivalent phase-changing screen (Rino, 2011). This approach abstracts the complex three-dimensional ionospheric structure into a statistically representative two-dimensional phase screen, providing the controlled and repeatable scintillation simulations essential for systematic error model development. The spectral density function (SDF) of the phase screen is modeled as a two-component power-law, defined as (Carrano & Rino, 2016):

$$\Phi_{\phi}(q) = C_p \begin{cases} q^{-p_1} & q \leq q_0 \\ q_0^{p_2-p_1} q^{-p_2} & q > q_0 \end{cases} \quad (1)$$

where C_p represents the turbulence strength, p_1 and p_2 are the spectral indices defining power-law slopes in different wavenumber regimes that determine the spectral shape characteristics, and q_0 is the break wavenumber which indicates the inner or outer scale of the irregularity structure and defines the transition point between the two spectral regimes (Carrano & Rino, 2016).

To achieve universal characterization independent of specific propagation geometries, the SDF can be simplified by scaling the wavenumber q with the Fresnel scale ρ_F to a normalized (and thus non-dimensional) unit such that $\mu = q\rho_F$. The Fresnel scale ρ_F represents the effective propagation distance from the phase screen defined by $\rho_F^2 = \lambda z / 2\pi$, at which diffractive effects become significant during wave propagation, where λ is the signal wavelength and z is the propagation distance. This normalization exploits the self-similar scaling properties of power-law processes, enabling universal characterization of scintillation effects independent of specific propagation geometries. Hence, the phase spectrum can be expressed in terms of the normalized wavenumber μ as (Carrano & Rino, 2016):

$$P(\mu) = \Phi_{\phi}(q) / \rho_F = \begin{cases} U|\mu|^{-p_1} & \mu \leq \mu_0 \\ U|\mu|^{-p_2} & \mu > \mu_0 \end{cases} \quad (2)$$

where the universal scattering strength U (i.e., the normalized phase spectral power at the Fresnel scale ρ_F) is defined as:

$$U = \begin{cases} C_p \rho_F^{p_1-1} & \mu_0 \geq 1 \\ C_p \rho_F^{p_1-1} \mu_0^{p_2-p_1} & \mu_0 < 1 \end{cases} \quad (3)$$

2.2 Phase Screen Generation and Wave Propagation

The computational implementation of the two-component power-law phase screen model requires systematic generation of phase screen realizations that are statistically equivalent to the specified spectral characteristics. The phase screen generation process begins with the creation of a one-dimensional random phase distribution based on the normalized spectral density function defined in Equation (2).

A statistically representative realization of the phase screen is generated by imposing the desired phase spectral density on white noise in the discrete Fourier transform domain. For a phase screen with N discrete points, the phase realization for the m^{th} Fourier component is constructed as (Rino et al., 2018):

$$\phi_{0,m} = \sum_{n=0}^{N-1} \sqrt{P(n\Delta\mu) \Delta\mu / (2\pi)} \eta_m e^{-2\pi i n m / N} \quad (4)$$

Here, η_m is a standard complex Gaussian random process with the white noise property whose variates are uncorrelated with unit-variance as $\langle \eta_m \eta_{m'}^* \rangle = \delta(m-m')$ and the Hermitian property $\eta_{N-m} = \eta_m^*$ which ensures purely real phase screen realizations (Rino, 2011).

Using the initial phase screen ϕ_0 , the propagation of the complex field at the Fresnel scale ρ_F is implemented through the following split-step algorithm based on the forward propagation equation under the parabolic approximation (Rino, 2011; Rino et al., 2018):

$$\hat{\psi}(0; n\Delta\mu) = \sum_{m=0}^{N-1} e^{i\phi_{0,m}} e^{-2\pi i n m / N} \quad (5)$$

$$\psi(\rho_F; m\Delta x / \rho_F) = \frac{1}{N} \sum_{n=0}^{N-1} \hat{\psi}(0; n\Delta\mu) e^{i(n\Delta\mu)^2 / 2} e^{2\pi i n m / N} \quad (6)$$

For time-series realization of scintillation, the spatial coordinate x is converted to temporal domain using the ratio ρ_F / v_{eff} :

$$\psi(t)\psi = \psi\left(\rho_F; t / \left(\rho_F / v_{\text{eff}}\right)\right) \quad (7)$$

In this conversion, t represents time, and v_{eff} is the effective scan velocity characterizing the relative motion between the propagation path and ionospheric irregularities, which is primarily determined by satellite motion and drift of irregularities projected onto the anisotropic irregularity plane.

The resulting phase ϕ and signal amplitude I are then obtained from the propagated complex wave field as:

$$\phi = \text{unwrap}\{\text{atan2}(\Re(\psi), \Im(\psi))\} \quad (8)$$

$$I = |\psi|^2 = \psi\psi^* \quad (9)$$

This propagation process enables the generation of realistic amplitude and phase time series that capture the scintillation effects arising from ionospheric irregularities. The space-to-time conversion through the parameter ρ_F / v_{eff} provides the temporal characteristics necessary for analyzing scintillation effects on GNSS measurements, with different values of ρ_F / v_{eff} generating varying temporal scales of the scintillation patterns.

2.3 Simulation Parameters and Configuration

The simulation systematically covers the parameter space relevant to equatorial scintillation conditions through controlled variation of the normalized spectral parameters $\{U, p_1, p_2, \mu_0\}$ defined in Section 2.1. The approach employs the space-to-time

conversion presented in Section 2.2 through the parameter ρ_F/v_{eff} to generate time-series realizations of scintillation effects across comprehensive parameter combinations.

Table 1 presents the parameter ranges used in this study. For the controlling parameters U and ρ_F/v_{eff} , we employ comprehensive ranges that encompass the variations observed in equatorial scintillation data analyzed in Xu et al. (2020) to ensure complete coverage of all potential scintillation conditions in equatorial regions.

Table 1. Parameter ranges and step sizes used for scintillation simulations

Parameter	Range	Step
U	0.1 ~ 3	0.1
p_1	2.7	-
p_2	3.6	-
μ_0	0.8	-
ρ_F/v_{eff}	0.1 ~ 3 (s)	0.1 (s)
C/N_0	30 ~ 55 (dB-Hz)	5 (dB-Hz)

The simulation parameters map to observable scintillation characteristics through two primary scintillation indices $\{S_4, \tau_0\}$. The scintillation index S_4 is defined as the normalized standard deviation of signal intensity fluctuations, typically calculated over 60-second intervals (Van Dierendonck et al., 1993). The decorrelation time τ_0 is defined as the time lag at which the intensity autocorrelation function decreases to $1/e$ of its maximum value, characterizing the temporal coherence of intensity fluctuations (Humphreys et al., 2009).

Numerical evaluation in Xu et al. (2020) demonstrates that the universal scattering strength U exhibits a direct one-to-one mapping relationship with the S_4 index. The decorrelation time τ_0 exhibits a linear relationship with ρ_F/v_{eff} , where its slope depends on the U value. As U increases, corresponding to stronger scintillation conditions, the slope decreases, indicating that under the same ρ_F/v_{eff} value, more severe scintillation results in faster time scales. The spectral shape parameters $\{p_1, p_2, \mu_0\}$ are fixed at their default values based on empirical observations of equatorial scintillation (Xu et al., 2020). This parametric approach enables systematic coverage of the $\{S_4, \tau_0\}$ parameter space required for comprehensive scintillation analysis.

Each parameter combination generates extensive time series data to ensure statistically significant characterization of scintillation effects. The simulation configuration includes a duration of 5×10^5 seconds per parameter set, sampled at 0.01-s intervals. The simulation outputs include user-end scintillation indices $\{S_4, \tau_0\}$ and phase transitions for both L1 and L5 frequencies. The dual-frequency implementation ensures that L1 and L5 signals experience the same ionospheric phase screen, with simulation parameters scaled according to frequency-dependent relationships while maintaining the same sequence of Gaussian noise for both frequencies. This approach preserves the physical correlation between frequencies while accounting for frequency-dependent scattering characteristics (Xu et al., 2020).

Figure 1 illustrates representative scintillation results from the simulation. The top panel shows the signal intensity fluctuations for L1 and L5 frequencies, with S_4 values of 0.97 and 1.09 and τ_0 values of 0.68 s and 0.57 s, respectively. The middle panel presents the initial refractive phase and the propagated phase that includes both refractive and diffractive effects for both frequencies. The bottom panel shows the extracted diffractive phase components, which clearly reveal the discrete phase transition events. The extraction process demonstrates how the initial phase screen ϕ_0 (representing purely refractive effects with only phase variations) differs from the propagated phase ϕ (which includes both refractive and diffractive effects). The diffractive phase term ϕ_d is obtained by differencing the propagated phase from the initial phase screen, $\phi_d = \phi - \phi_0$, effectively isolating the diffractive scintillation effects responsible for the discrete phase transition events observed in GNSS signals (Breitsch et al., 2020).

In real GNSS observations, scintillation effects on the receiver are significantly dependent on the noise characteristics of the signal. We consider a baseline carrier-to-noise density ratio (C/N_0) as the relevant parameter determining the impact of thermal noise on the receiver, and it is primarily determined by satellite elevation angle. The simulation incorporates the C/N_0 ranges and step sizes given in Table 1, with L5 signals assigned values 3.5 dB-Hz higher than corresponding L1 values based on empirical measurements of received power (Hrbek et al., 2016). For a given C/N_0 , the noise imposed on the phase measurements can be derived as:

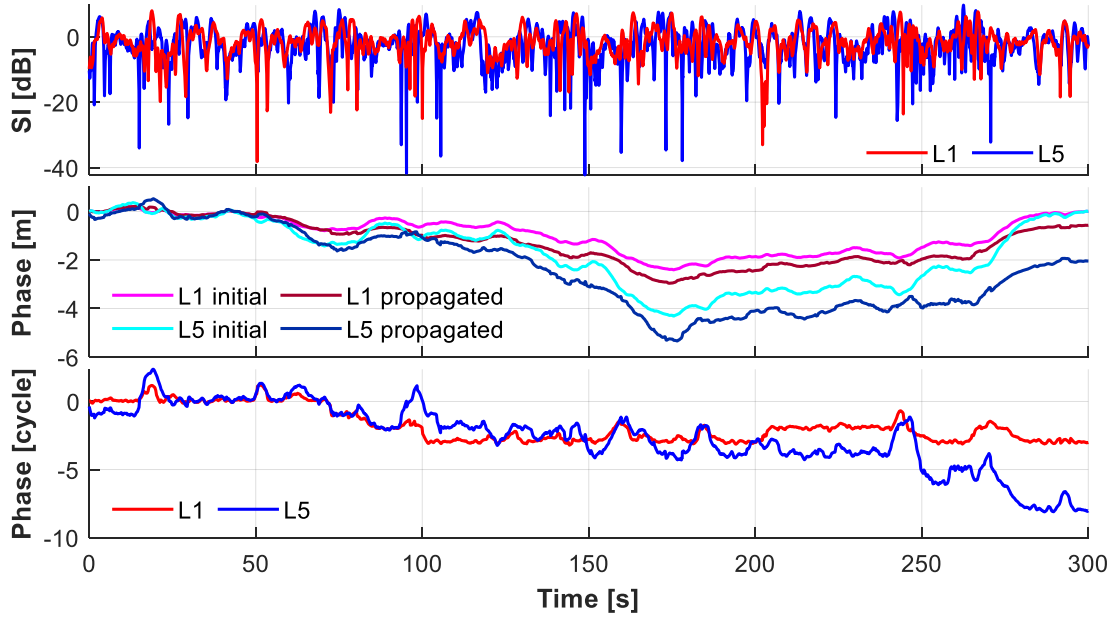


Figure 1. Scintillation simulation output for parameter set $\{U, \rho_F/v_{\text{eff}}\} = \{3, 1.2\}$: (top) signal intensity fluctuations for L1 and L5 frequencies, (middle) initial refractive phase and propagated phase including diffractive effects for both frequencies, and (bottom) extracted diffractive phase components obtained by differencing the propagated and initial phases.

$$n = \sigma\eta, \quad \sigma = \sqrt{\frac{B}{C/N_0}} \quad (10)$$

Here, η is a standard complex Gaussian random variable with real and imaginary components each having variance $\sigma^2/2$ where σ is the noise standard deviation. The noise bandwidth B is assumed to be $1/T$ where T is the integration time (i.e., sampling interval of raw measurements which is 0.01 s for this study) (Breitsch et al., 2020).

The resulting simulation approach enables systematic generation of scintillation scenarios across the complete parameter space required for GBAS error modeling. Each parameter combination produces statistically consistent occurrence processes of phase transitions that reliably represent equatorial ionospheric conditions, providing the controlled foundation necessary for the stochastic modeling approach presented in the following section.

III. MODEL-BASED OVERBOUNDING METHODOLOGY

While the physics-based simulation framework enables controlled generation of scintillation scenarios, deriving conservative error bounds (σ_{scint}) at GBAS integrity probability levels (10^{-7} to 10^{-9}) requires addressing the fundamental limitations of conventional statistical approaches when applied to scintillation-induced errors. This section develops a novel stochastic modeling methodology that represents phase transitions as Skellam-distributed point processes, enabling analytical extrapolation to rare-event probability levels without requiring prohibitively large sample sizes.

3.1 Limitations in Conventional Error Bounding

The conventional approach widely used for error bounding in GNSS integrity applications is CDF overbounding (DeCleeve, 2000; Blanch et al., 2018). CDF overbounding inflates the standard deviation of a Gaussian distribution to ensure it conservatively bounds the CDF derived from finite data samples. The inflated Gaussian sigma is determined by finding the minimum value that satisfies the overbounding constraint across the entire error distribution.

However, during scintillation, phase transitions manifest as impulse-like disturbances that propagate through the carrier smoothing filter, where each discrete transition introduces a step input that generates transient errors decaying exponentially. The magnitude and decay time of these transient errors depend critically on the smoothing filter's impulse response, where longer time constants result in slower decay rates and extended decay times. When multiple phase transitions occur before complete

reconvergence from previous transitions, the resulting errors superpose additively, resulting in cumulative residual errors. This error accumulation generates distributions with significantly heavier tails than typical measurement errors. These phase transition-induced errors can exhibit extreme outliers that, despite being rare events, become critically important for aviation safety at integrity probability levels down to 10^{-9} . Direct CDF overbounding on samples cannot reliably characterize these extreme tail behaviors due to finite sample size limitations. This necessitates alternative analytic approaches that capture the underlying stochastic processes causing these errors.

3.2 Stochastic Modeling of Phase Transitions

To overcome the limitations of direct CDF overbounding, this research develops a novel stochastic modeling approach that represents phase transitions as point processes. Rather than relying solely on finite sample statistics, this methodology models the underlying physical mechanisms causing phase transition-induced errors, enabling analytical derivation of probability distributions and statistical extrapolation to rare-event probability levels required for aviation integrity. The approach comprises three key components: systematic detection and quantization of phase transitions from simulated data, representation of transition events as Poisson point processes following Skellam distributions, and incorporation of correlation structures observed in scintillation phenomena.

3.2.1 Detection and Quantization of Phase Transitions

Phase transition detection requires systematic identification and quantization from simulated diffractive phase time series. We employ Total Variation Denoising (TVD), an optimization technique for estimating signals with sparse derivative components in the presence of noise (Breitsch et al., 2020; Rudin et al., 1992; Selesnick, 2014). Since direct rounding of diffractive phase may count noise-induced fluctuations as false transitions, we implement signal intensity weighting based on the physical characteristic that phase transitions occur during deep signal fading, referred to as canonical fading in the literature (Humphreys et al., 2010). The weighted TVD cost function applied to diffractive phase ϕ_d becomes:

$$J = \min_u \left\{ \frac{1}{2} \|\phi_q - \phi_d\|_2^2 + \sum_m \alpha_m |\phi_{q,m+1} - \phi_{q,m}| \right\} \quad (11)$$

where ϕ_q is the quantized output and α_m is the regularization parameter that penalizes magnitude changes (10 during signal fading < -5 dB, 1 otherwise). Individual phase transitions are identified by rounding the TVD result to nearest integers and detecting non-zero differences.

3.2.2 Poisson Process Representation

Based on the modeling approach presented by Breitsch et al. (2020), phase transitions are modeled as Poisson point processes. The positive and negative phase transitions, $N^{(+/-)}$, are represented as independent Poisson processes:

$$N^{(+)} \sim \text{Poisson}(\lambda_+), \quad N^{(-)} \sim \text{Poisson}(\lambda_-) \quad (12)$$

The net signed transitions over each time window follows a Skellam distribution:

$$X = N^{(+)} - N^{(-)} \sim \text{Skellam}(\lambda_+, \lambda_-) \quad (13)$$

Extensive simulation analysis reveals approximately equal occurrence rates for positive and negative transitions, leading to a symmetric Skellam distribution where $\lambda_+ = \lambda_- = \lambda$, with mean $E[X] = 0$ and variance $\text{Var}[X] = 2\lambda$ (Breitsch et al., 2020).

Figure 2 illustrates the phase transition detection and quantization process applied to simulated diffractive phase data. The top panel demonstrates how the TVD algorithm effectively captures discrete phase jumps, with the quantized fit (black dashed line) clearly identifies step-like transitions. The middle panel shows the cumulative occurrence of positive and negative transitions, revealing the stochastic nature of phase transition events. The bottom panel presents the net transition count, representing the Skellam-distributed process used for stochastic modeling.

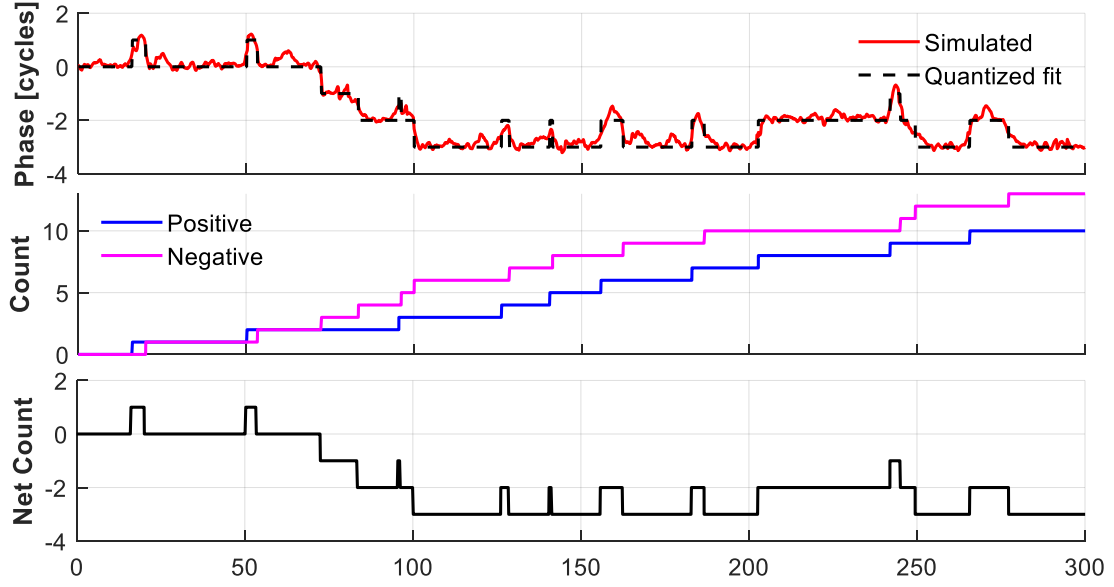


Figure 2. Phase transition detection and quantization using weighted TVD on the simulated results with $\{U, \rho_F/v_{\text{eff}}\} = \{3, 1.2\}$. (Top) Simulated diffractive phase (red) and quantized fit (black dashed) showing discrete phase transitions. (Middle) Cumulative counts of positive (blue) and negative (magenta) transitions. (Bottom) Net transition count representing the Skellam-distributed process used for stochastic modeling.

3.2.3 Correlation Structure Modeling

The fundamental Poisson process representation provides the foundation for incorporating correlation structures that capture the complex dependencies observed in scintillation data. Real scintillation data exhibits correlation patterns both within (intra-frequency) and across frequencies (inter-frequency) that must be incorporated into the stochastic model. The decomposition property of Poisson processes enables modeling these correlations through additive components (Gallager, 1996). For intra-frequency correlations, we decompose the Poisson processes as:

$$N^{(+)} = U + V^{(+)}, \quad N^{(-)} = U + V^{(-)} \quad (14)$$

where $U \sim \text{Poisson}(\lambda_u)$ represents the correlated source between positive and negative transitions, and $V_{+/-} \sim \text{Poisson}(\lambda_v)$ are independent sign-specific components. This decomposition yields:

$$X = N^{(+)} - N^{(-)} = V^{(+)} - V^{(-)} \sim \text{Skellam}(\lambda_v) \quad (15)$$

For dual-frequency applications, inter-frequency correlations are modeled through extended decomposition into frequency-correlated, sign-correlated, and frequency-specific components:

$$N_1^{(+/-)} = U_C + U_1 + V_C^{(+/-)} + V_1^{(+/-)}, \quad N_5^{(+/-)} = U_C + U_5 + V_C^{(+/-)} + V_5^{(+/-)} \quad (16)$$

where U_C represents the common correlated source across all frequencies and signs, U_1 and U_5 are frequency-specific correlated sources, $V_C^{(+/-)}$ are sign-specific correlated sources across frequencies, and $V_1^{(+/-)}$ and $V_5^{(+/-)}$ are frequency-specific and sign-specific independent components. The resulting Skellam distributions for each frequency become:

$$X_1 = V_C^{(+)} + V_1^{(+)} - V_C^{(-)} - V_1^{(-)} \sim \text{Skellam}(\lambda_{v,C} + \lambda_{v,1}) \quad (17)$$

$$X_5 = V_C^{(+)} + V_5^{(+)} - V_C^{(-)} - V_5^{(-)} \sim \text{Skellam}(\lambda_{v,C} + \lambda_{v,5}) \quad (18)$$

We extract the transition rate parameters $\lambda_{v,C}$, $\lambda_{v,1}$, and $\lambda_{v,5}$ from the process covariances and variances. Since transition rates are sensitive to the counting time window, we apply adaptive time window, varying with scintillation conditions. This decomposition accurately represent the complex correlations observed in scintillation data while maintaining analytic tractability required for statistical error bounding.

3.3 Derivation of Smoothed Errors

3.3.1 Hatch Filter Response to Phase Transitions

The GBAS carrier smoothing filter implements exponentially weighted averaging through the recursive formulation:

$$\bar{\rho}_k = (1 - \beta)\rho_k + \beta(\bar{\rho}_{k-1} + \phi_k - \phi_{k-1}) \quad (19)$$

where τ is the smoothing time constant, Δt is the sample interval, and $\beta = 1 - \Delta t/\tau$ represents the smoothing coefficient. ρ_k is the pseudorange, ϕ_k is the carrier phase, and $\bar{\rho}_k$ is the smoothed pseudorange at epoch k .

To analyze phase transition effects, we consider error propagation assuming zero pseudorange input ($\rho_k = 0$) and define the incremental phase input $\Delta\phi_k = \phi_k - \phi_{k-1}$. The smoothed error due to phase transitions then becomes:

$$e_k = \beta(e_{k-1} + \Delta\phi_k) = \sum_{m=1}^k \beta^m \Delta\phi_{k-m+1} \quad (20)$$

3.3.2 Convolution Formulation

The filtered output can be expressed as a convolution of the filter impulse response with phase transition increments:

$$S = \sum_{n=1}^N h[n] \cdot \Delta\phi_{k-n} = \sum_{n=1}^N h[n] \cdot X_n \quad (21)$$

where $h[n] = \beta_n$ represents the filter impulse response and X_n denotes phase transition increments following the Skellam distribution $X_n \sim \text{Skellam}(\lambda\Delta t)$. The smoothed error results from this weight accumulation of Skellam-distributed increments, whose statistical distribution can be derived analytically.

For dual-frequency smoothing modes (DFree and IFree), the smoothed error expression extends to:

$$S = \sum_{n=1}^N h[n] \cdot (a_1 X_n^{(1)} + a_5 X_n^{(5)}) \quad (22)$$

where a_1 and a_5 represent the combination weights for different smoothing modes as given in Table 2. $X_n^{(1)}$ and $X_n^{(5)}$ denote the phase transition processes for L1 and L5 frequencies, respectively.

Table 2. Combination weights for smoothing modes, where $\gamma = f_{L1}^2/f_{L5}^2$ represents the squared frequency ratio

Smoothing mode	a_1	a_5
L1 Single-Frequency	λ_{L1}	0
L5 Single-Frequency	0	λ_{L5}
Divergence-Free	$\lambda_{L1} \cdot (\gamma + 1) / (\gamma - 1)$	$-2\lambda_{L5} / (\gamma - 1)$
Ionosphere-Free	$\lambda_{L1} \cdot \gamma / (\gamma - 1)$	$-\lambda_{L5} / (\gamma - 1)$

3.4 Statistical Error Bounding

3.4.1 Skellam-Based Quantile Estimation

The stochastic modeling approach enables analytical tail estimation through the cumulant generating function (CGF) of the Skellam distribution:

$$K_X(s) = 2\lambda\Delta t(\cosh(s) - 1) \quad (23)$$

For the weighted sum of independent Skellam variables representing the smoothed error:

$$K_S(s) = \sum_{n=1}^N K_X(h[n] \cdot s) = 2\lambda\Delta t \sum_{n=1}^N (\cosh(h[n] \cdot s) - 1) \quad (24)$$

For dual-frequency smoothing modes, the combined CGF incorporates both L1 and L5 phase transition processes using the combination weights defined in Table 2:

$$K_S(s) = 2\lambda_1\Delta t \sum_{n=1}^N (\cosh(a_1 h[n] \cdot s) - 1) + 2\lambda_5\Delta t \sum_{n=1}^N (\cosh(a_5 h[n] \cdot s) - 1) \quad (25)$$

where a_1 and a_5 are the combination weights for the specific smoothing mode, and λ_1 and λ_5 are the phase transition rates for L1 and L5 frequencies, respectively. This formulation enables derivation of conservative error bounds for all GBAS smoothing modes under scintillation conditions.

For a target tail probability p , we determine the quantile such that $P(S > q) = p$ using the saddlepoint approximation (Lugannani & Rice, 1980). This requires solving for the saddlepoint s^* where $K_S(s^*) = q$, computing auxiliary variables:

$$w = \sqrt{2(s^*q - K_S(s^*))}, \quad u = s^* \sqrt{K_S''(s^*)} \quad (26)$$

and applying the Lugannani-Rice formula (Lugannani & Rice, 1980):

$$P(S > q) = 1 - \Phi(w)\varphi(w) + (1/w - 1/u) \quad (27)$$

where Φ and φ represent the standard normal CDF and PDF, respectively.

3.4.2 Gaussian Overbounding

The saddlepoint approximation provides quantile-probability pairs (q, p) for the Skellam-distributed smoothed error, but GBAS error modeling requires conservative Gaussian sigma values. To integrate the Skellam-based error bounds into GBAS operations, we apply CDF overbounding by finding an inflated Gaussian sigma that convers analytically derived distribution of phase transition-induced errors.

We determine the equivalent Gaussian sigma $\sigma_{ovbd,m}$ that ensures the Gaussian distribution covers all saddlepoint-derived (q, p) pairs:

$$\sigma_{ovbd,m} = \max_i \{q_i / \Phi^{-1}(1 - p_i)\} \quad (28)$$

where (q_i, p_i) represents the quantile-probability pairs obtained from saddlepoint approximation across target probabilities from 10^{-4} to 10^{-9} , and Φ^{-1} is the inverse standard normal CDF.

Figure 3 demonstrates the fundamental advantage of the model-based approach over conventional sample-based overbounding. The left panels show phase transitions and their propagation through the 100-second Hatch filter, generating the smoothed error time series. The right panel compares distribution tails: the empirical distribution (black solid) represents the sample-based complementary CDF from simulation data, while the conventional sample-based overbound $\sigma_{ovbd,s}$ (blue dashed) is derived directly from finite sample statistics following the standard GBAS approach.

However, the conventional sample-based approach fails to provide reliable estimates at aviation integrity probability levels (especially, 10^{-7} to 10^{-9}) with resolution constrained by finite sample size. In contrast, the model-based approach uses Skellam quantiles (magenta dots) to analytically characterize these extreme probability levels, capturing the underlying stochastic process rather than relying on finite samples. The resulting model-based Gaussian overbound $\sigma_{ovbd,m}$ (red dashed) ensures conservative coverage of all analytically-derived quantile-probability pairs, providing statistically significant bounds at the rare-event probability levels required for aviation integrity applications.

This model-based methodology represents a fundamental advancement over conventional sample-based overbounding by modeling the underlying stochastic processes causing the errors, enabling reliable extrapolation to probability levels that are otherwise inaccessible through direct sampling approaches.

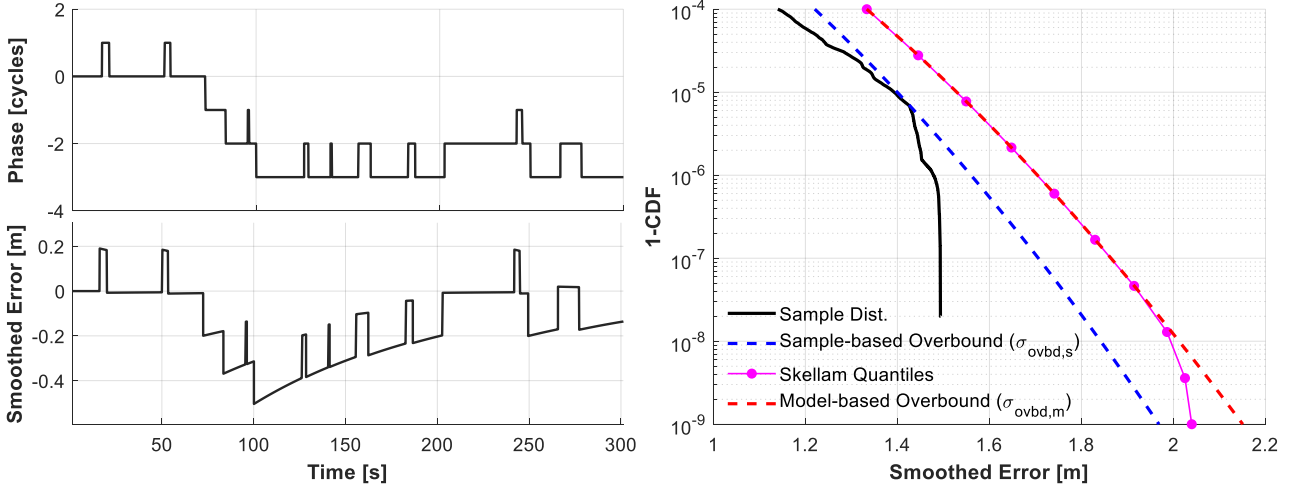


Figure 3. Comparison of sample-based and model-based Gaussian overbounding approaches for phase transition-induced smoothed errors. (Left) Time series showing simulated phase transitions with $\{U, \rho_F/v_{\text{eff}}\} = \{3, 1.2\}$ and resulting smoothed errors with 100-second smoothing. (Right) Complementary CDF comparing empirical distribution (black solid), conventional sample-based overbound (blue dashed), Skellam-based quantiles (magenta dots), and model-based overbound (red dashed).

IV. RESULTS AND DISCUSSION

4.1 Sample-Based and Model-Based Overbounding Comparison

Figure 4 compares sigma overbounds (σ_{ovbd}) derived from sample-based and model-based approaches for single-frequency L1 smoothing with 100-second time constant at $C/N_0 = 55$ dB-Hz as a function of S_4 and τ_0 at L1 frequency. Both approaches demonstrate that σ_{ovbd} increases with stronger scintillation (higher S_4) and faster fading (shorter τ_0), consistent with more frequent signal fades and phase transitions under severe scintillation conditions (Sun et al., 2025a).

The model-based approach demonstrates superior smoothness and consistency compared to sample-based methods. While sample-based results (panel a) exhibit irregular variations at moderate to high scintillation levels (L1 $S_4 > 0.6$), the model-based methodology (panel b) produces smooth, continuous error bounds across the entire parameter space. The cross-sectional analysis (panels c and d) reveals that the model-based approach consistently provides higher, more conservative bounds, particularly under strong scintillation (high S_4) and fast fading conditions (shorter τ_0) where phase transition dynamics dominate. This conservative behavior addresses fundamental limitations of empirical overbounding methods due to finite sample limitations when applied to rare-event probability analysis required for aviation integrity applications.

4.2 Dual-Frequency Smoothing Mode Error Bounds

Figure 5 presents model-based sigma overbounds for dual-frequency smoothing modes, revealing that dual-frequency combination fundamentally amplify rather than mitigate phase transition-induced errors. IFree smoothing with a 100-second time constant produces sigma overbounds of approximately 3.4 m at the worst case, driven by the combination coefficients $\gamma/(\gamma - 1) \approx 2.26$ for L1 phase and $1/(\gamma - 1) \approx 1.26$ for L5 phase, where $\gamma = f_{L1}^2/f_{L5}^2 \approx 1.79$. DFree smoothing exhibits even more severe amplification, with values reaching 6.3 m at the worst case due to the larger combination coefficients: $(\gamma + 1)/(\gamma - 1) \approx 3.52$ for L1 phase and $2/(\gamma - 1) \approx 2.52$ for L5 phase. DFree coefficients for both L1 and L5 phases are larger than IFree coefficients, indicating that DFree may be more susceptible to phase transition effects under scintillation. These findings indicate that planned dual-frequency GBAS architectures that specify DFree smoothing as the primary mode may require further investigation (Murphy et al., 2021).

The cross-sectional analysis (panels c and d) establishes a clear performance hierarchy: L1 < L5 < IFree < DFree. Panel c shows the amplification becomes pronounced at moderate scintillation levels (L1 $S_4 > 0.5$), with DFree L1 exceeding L1-only errors by approximately 7 times at strong scintillation. This amplification occurs because, unlike first-order refractive delays that scale with frequency and can be mostly eliminated through linear combinations, diffractive phase transitions occur as discrete events that are temporally decorrelated between frequencies (Carrano et al., 2012; Jiao et al., 2016). The dual-frequency combination coefficients enhance these decorrelated errors rather than canceling them, making dual-frequency operations most susceptible to phase transition effects during scintillation.

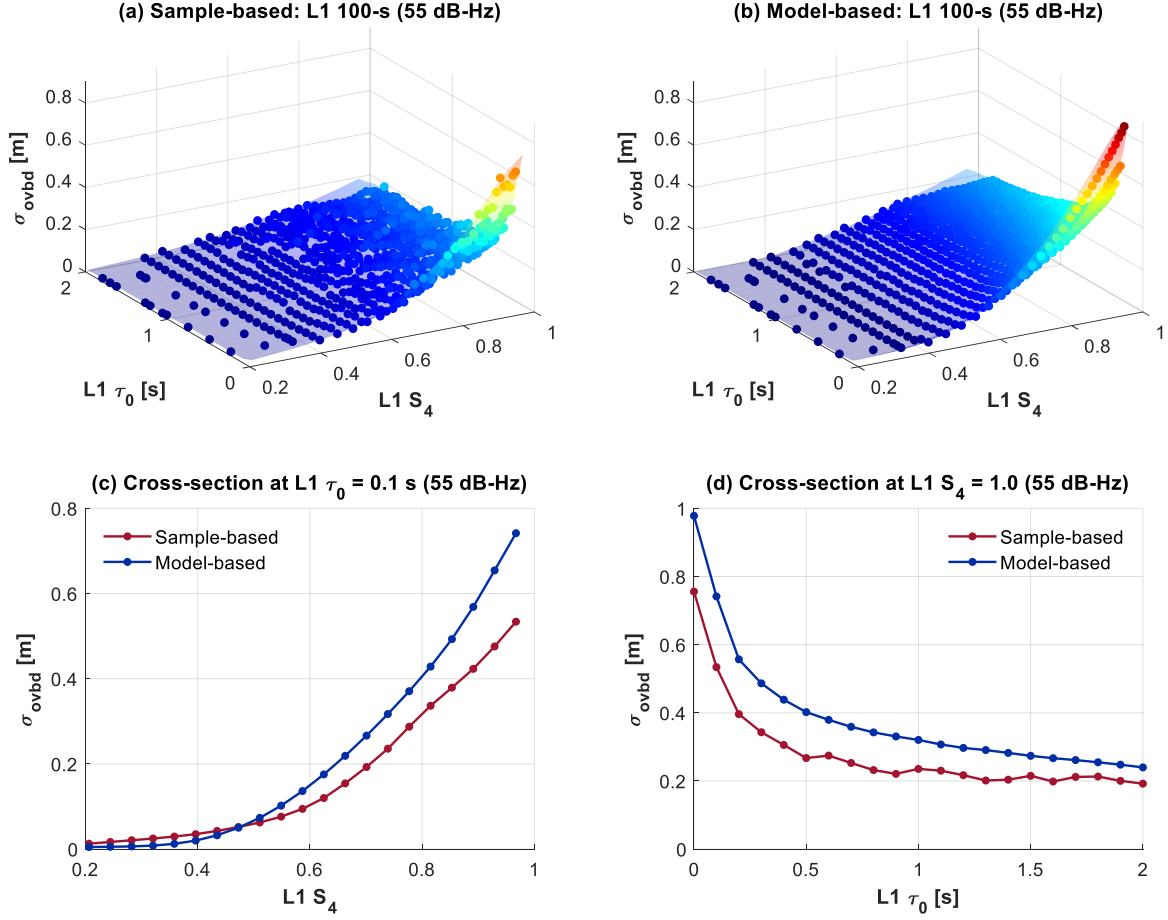


Figure 4. Comparison of sample-based and model-based sigma overbounds for single-frequency L1 smoothing modes (100-s smoothing, $C/N_0 = 55$ dB-Hz). (a) Sample-based L1 σ_{ovbd} contours derived from empirical CDF overbounding, (b) Model-based σ_{ovbd} contours derived from stochastic point process modeling, (c) Cross-section at $\tau_0 = 0.1$ s showing method comparison along S_4 , (d) Cross-section at $S_4 = 1.0$ showing method comparison along τ_0 .

4.3 Smoothing Time Constant Effects

Figure 6 demonstrates the critical impact of extended smoothing time constants on scintillation-induced error bounds for DFree smoothing mode. This analysis is particularly relevant for dual-frequency GBAS architectures where DFree L1 with 600-second time constants is planned as the primary operational mode due to its ability to remove ionospheric divergence effects and achieve lower noise levels through extended averaging (Murphy et al., 2021).

A 600-s smoothing time constant results in approximately $2.4\times$ larger errors compared to a 100-s time constant across all scintillation conditions. This amplification occurs because each discrete phase transition introduces a persistent error that decays exponentially with the smoothing time constant τ . Under scintillation conditions where multiple phase transitions occur within the smoothing window, longer filters cannot recover between successive events, leading to accumulated residual errors that compound the overall error magnitudes. This finding highlights a fundamental trade-off between noise reduction benefits and scintillation vulnerability in extended smoothing implementations for equatorial GBAS operations.

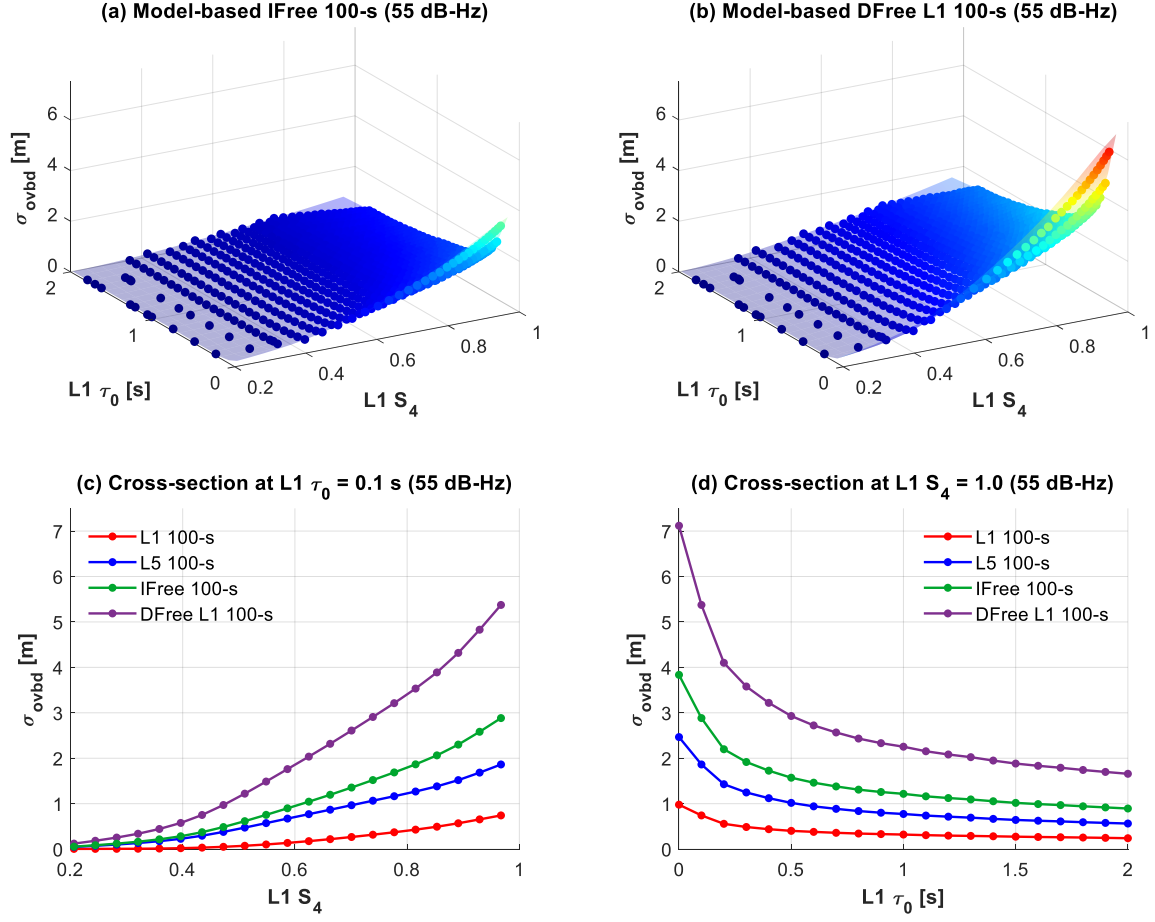


Figure 5. Dual-frequency mode error amplification using model-based approach (100-s smoothing, $C/N_0 = 55$ dB-Hz). (a) IFree σ_{ovbd} contours. (b) DFree L1 σ_{ovbd} contours. (c) Cross-section at $\tau_0 = 0.1$ s comparing all smoothing modes (L1, L5, IFree, and DFree L1) along S_4 , (d) Cross-section at $S_4 = 1.0$ comparing all smoothing modes (L1, L5, IFree, and DFree L1) along τ_0 .

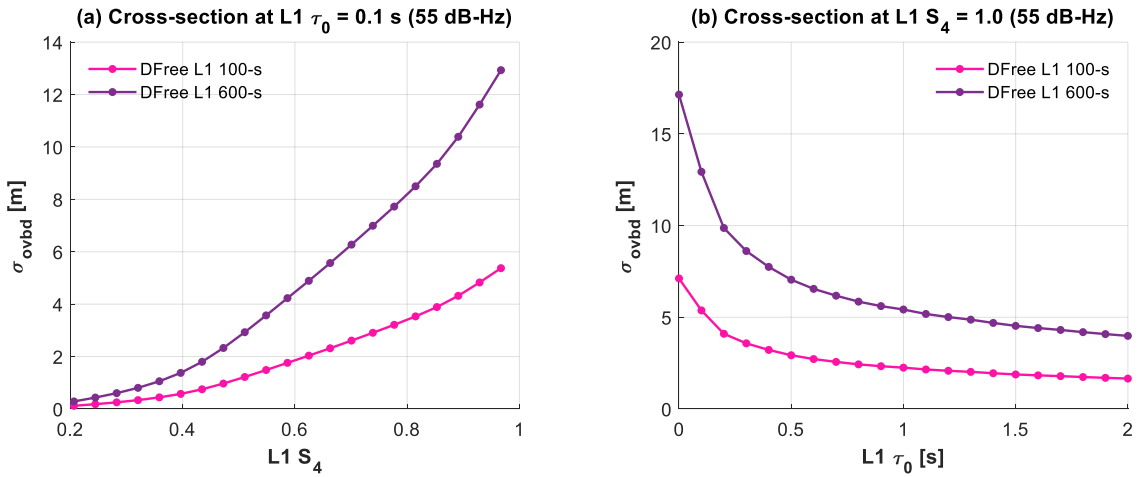


Figure 6. Smoothing time constant effects on DFree L1 error bounds using model-based approach ($C/N_0 = 55$ dB-Hz). (a) Cross-section at $\tau_0 = 0.1$ s comparing 100-s and 600-s time constants along S_4 , (d) Cross-section at $S_4 = 1.0$ comparing time constants along τ_0 .

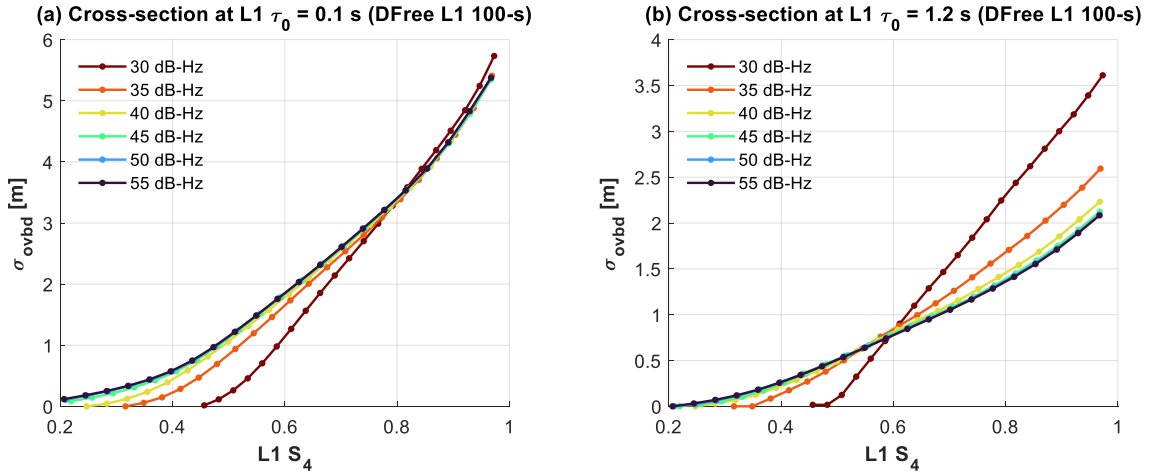


Figure 7. Effect of baseline C/N_0 on DFree L1 error bounds using model-based approach (100-s smoothing). (a) Cross-section at $\tau_0 = 0.1$ s comparing multiple C/N_0 levels along S_4 . (b) Cross-section at $\tau_0 = 1.2$ s comparing multiple C/N_0 levels along S_4 .

4.4 Baseline Carrier-to-Noise Ratio Effects

Figure 7 examines the impact of baseline C/N_0 on sigma overbounds for DFree 100-second smoothing, revealing that baseline C/N_0 effects vary significantly depending on the scintillation temporal characteristics. Given that baseline C/N_0 varies with satellite elevation angle, these results illustrate how elevation-dependent effects interact with scintillation parameters.

The results demonstrate that C/N_0 effects vary significantly depending on the scintillation time scale, with distinct behavior patterns emerging between fast ($\tau_0 = 0.1$ s) and slow ($\tau_0 = 1.2$ s) fading conditions. Under fast fading ($\tau_0 = 0.1$ s, panel a), minimal variations occur between different C/N_0 levels across most S_4 values, with curves overlapping until very strong scintillation ($S_4 > 0.8$). This behavior occurs because phase transition effects dominate under fast fading conditions, making the scintillation-induced errors relatively insensitive to thermal noise variations.

Under slower fading conditions ($\tau_0 = 1.2$ s, panel b), C/N_0 effects become pronounced across all S_4 levels. Lower C/N_0 conditions show dramatically increased sigma overbounds, with the 30 dB-Hz case reaching 3.6 m compared to 2.2 m for the 55 dB-Hz case at $S_4 = 1.0$. This amplification occurs because thermal noise due to lower baseline C/N_0 becomes a significant contributor under slower fading conditions, where increased noise induces additional phase transitions (or "noise-induced cycle slips" as termed in Breitsch et al., 2020), effectively increasing the total phase transition rate when combined with scintillation-induced transitions.

V. CONCLUSIONS

This paper presented physics-based scintillation simulation and a Skellam-based stochastic modeling approach to derive sigma overbounds for phase transition-induced errors in GBAS carrier-smoothed pseudoranges under ionospheric scintillation. The physics-based simulation enables controlled generation of scintillation scenarios across comprehensive parameter spaces, while the stochastic modeling enables statistical extrapolation to aviation integrity risk levels (10^{-7} to 10^{-9} per operation) that conventional empirical methods cannot reliably achieve. The proposed approach derived sigma overbounds that account for phase transition effects across scintillation conditions for L1 and L5 single-frequency smoothing, IFree, and DFree modes.

The results indicate that dual-frequency smoothing modes can amplify phase transition-induced errors, with the derived overbounds revealing a clear order of error levels: $L1 < L5 < \text{IFree} < \text{DFree}$. Although the IFree and DFree smoothing modes are designed to mitigate first-order ionospheric delays and ionospheric divergence effects, respectively, both modes amplify phase transition-induced errors instead. This amplification results from their linear combination coefficients, which scale and combine the phase transition-induced errors from the L1 and L5 signals. IFree smoothing mode results in errors approximately 3.5 times larger and DFree results in errors 7 times larger than L1 single-frequency smoothing mode across all simulated scintillation conditions.

The analysis of smoothing time constant effects reveals additional consideration for GBAS operations under scintillation. While extended time constants effectively reduce noise levels, they systematically amplify phase transition-induced errors by a factor

of 2.4 for 600-s smoothing compared to 100-s across all scintillation conditions. The baseline C/N_0 analysis demonstrates additional variability, where thermal noise interactions vary significantly between fast and slow fading conditions, with pronounced amplification under slower fading conditions.

The derived sigma overbound parameters provide the foundation for incorporating phase transition effects into GBAS protection level calculations during scintillation. This enables comprehensive performance evaluation and supports development of operational strategies such as adaptive smoothing mode selection and optimized time constant configuration. Future work should investigate residual phase transition-induced errors that remain after implementing existing GBAS integrity monitors and develop real-time scintillation monitoring to ensure reliable GBAS performance under scintillation conditions.

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